VARUN PRASAD S

RTL Design & Verification Engineer at Kiwistron

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CAREER OBJECTIVE

Entry-level VLSI Design & Verification Engineer with strong hands-on experience in RTL design using Verilog/SystemVerilog and protocol-based design. Proficient in FSM-based control logic, combinational and sequential circuit design, and simulation-driven debugging. Also brings practical embedded systems, IoT, and PCB design experience from professional project work. Seeking an opportunity to contribute to RTL development and functional verification in a semiconductor design environment.

PROFESSIONAL EXPERIENCE

RTL Design & Verification Engineer – Kiwistron **Feb 2026 – Present**

- Design and verify RTL modules in Verilog, running simulations using Verilator.
- Use VS Code as the primary environment for RTL development and debugging.
- Synthesize RTL designs using Yosys.
- Perform Physical Design (PD) tasks using OpenLane, OpenROAD, Magic and KLayout, covering synthesis, floorplanning, and layout verification.
- Inspect final GDSII layouts using KLayout and a GDS 3D viewer.
- Work within a Linux-based EDA tool environment across the RTL-to-PD design flow.

TRAINING & CERTIFICATION

- Advanced VLSI Design & Verification – Maven Silicon Softech Pvt. Ltd
- PMKVY Certification – VLSI Engineer – Electronics Sector Skill Council of India

PROJECTS

Approximate MAC Unit for Image Processing

- Designed and implemented an approximate Multiply–Accumulate (MAC) unit using Verilog targeting image-processing applications.
- Integrated Roba Multiplier (2016) architecture to reduce area and power with acceptable error metrics.
- Verified functionality using custom testbenches and waveform analysis.
- Synthesised and simulated using Vivado and Xilinx ISE.

AHB to APB Bridge Design

- Designed RTL for AMBA AHB-to-APB bridge supporting high-speed to low-power peripheral communication.
- Implemented FSM-based control logic for protocol translation.
- Verified read/write transactions through simulation and debugged timing issues.
- Developed using Verilog and simulated in Xilinx ISE.

1×3 Router Design

- Designed a 1×3 packet router with address-based data routing.
- Implemented arbitration and control logic using Verilog.
- Verified functionality through RTL simulation in Vivado.

Additional projects available on GitHub – see contact details above

PROFESSIONAL PROJECTS

Companion Robot – AI-Based, PCB & Embedded Systems

- Designed a complete PCB using KiCad and developed embedded firmware on ESP32.
- Integrated OLED display, capacitive touch sensor and buzzer.
- Integrated AI models using Groq and ElevenLabs for intelligent interaction.
- Designed animated OLED eyes for interaction; touch sensor input; date and time display with alarm.
- Built a dedicated website to change the robot's eyes and set timers.
- Designed a compact PCB to avoid wiring and reduce the overall size of the robot.
- Architected modular embedded software for scalable feature integration.
- Performed PCB testing, debugging, and hardware validation.

AIoT & IoT Development

- Developed IoT solutions using ESP32 and ESP8266 for Wi-Fi based monitoring and control systems.
- Designed real-time dashboards for sensor data visualization and monitoring.
- Built AIoT-based automation and smart monitoring applications with cloud communication.

Robotics Projects

- Developed a Companion Robot integrating embedded firmware with sensors and actuators.

- Contributed to hardware integration for a Spider Robot platform.
- Debugged hardware and firmware issues during testing and validation.

CAD & Product Development

- Used CorelDRAW to design a car chassis and simple 2D products for product development.

Technical Documentation

- Authored technical documentation for embedded and AIoT products, including user guides, API documentation, and hardware setup manuals.
- Produced technical blogs and product documentation to support user adoption.

Additional projects available on GitHub – see contact details above

EDUCATION

- Bachelor of Engineering (B.E.) – Electronics and Communication Engineering, K. Ramakrishnan College of Engineering, Tiruchirappalli (Autonomous) CGPA: 7.92/10 2025
- Higher Secondary Certificate (HSC), Eden Gardens Matric Higher Secondary School 74.7% 2021
- Secondary School Leaving Certificate (SSLC), Annai Paruvathamma Matriculation School 66.6% 2019

TECHNICAL SKILLS

- **HDL:** Verilog, SystemVerilog
- **RTL Design:** FSM (Moore & Mealy), combinational & sequential logic, pipelining basics, blocking vs non-blocking assignments, reset strategies
- **Design & Verification:** VS Code, Verilator, Yosys; directed testbenches, waveform-based debugging
- **Physical Design (PD):** OpenLane, KLayout, OpenROAD, Magic, GDS 3D Viewer
- **Protocols:** AMBA AHB, APB, SPI, USART, I2C
- **Embedded Systems:** ESP32, ESP8266, Arduino, STM32 Blue Pill, Embedded C, UART, I2C, SPI, GPIO, PWM
- **PCB Design:** KiCad, Fritzing, Schematic Capture, PCB Layout, Gerber Generation, DRC/ERC, Hardware Debugging
- **IoT & AIoT:** ESP32, MQTT, Wi-Fi, Sensor Integration, Dashboard Development, Groq and ElevenLabs APIs; Python-based anomaly detection; AI camera applications for gesture control, face recognition and fire detection
- **EDA Tools:** Vivado, Xilinx ISE, Quartus Prime, Cadence Virtuoso (basic)
- **Software:** CorelDRAW, Fritzing, Arduino IDE
- **Other Tools:** KiCad, Git, GitHub, VS Code
- **Programming:** C, C++, Python
- **OS:** Linux, Windows

INTERNSHIP

- Verilog Design Intern – Maven Silicon
- Embedded System Internship

ACHIEVEMENTS

- Served as Class Representative for four academic years, coordinating between students and faculty.
- Organised technical idea-presentation sessions to promote peer learning and technical discussion.
- Coordinated department-level technical symposium events with high student participation.
- Demonstrated leadership and teamwork in managing academic and technical activities.

WORKSHOPS & TECHNICAL EXPOSURE

- Attended IoT & Devices Workshop conducted by IIT Madras, covering IoT system architecture and devices.
- Participated in an Embedded Systems Workshop focusing on microcontroller-based system fundamentals.
- Completed a Web Development Workshop covering basic front-end application concepts.
- Attended an Innovative Electronics Workshop exploring practical electronics design approaches.

HOBBIES

- Playing cricket • Video editing and basic motion graphics • Casual gaming

DECLARATION

I hereby declare that the information provided above is true to the best of my knowledge.